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Enhancing healthcare utilization and reducing preventable hospitalizations: exploring the healthcare-seeking propensity of patients with non-communicable diseases in Rural China

Yanqiu Hou[1], Wenyu Li[1], Shangai Li[2], Linxuan Chen[1], Jiayu An[1] and Shan Lu[1,3*]

Abstract

Objective Understanding healthcare-seeking propensity is crucial for optimizing healthcare utilization, especially for patients with chronic conditions like hypertension or diabetes, given their substantial burden on healthcare systems globally. This study aims to evaluate hypertensive or diabetic patients' healthcare-seeking propensity based on the severity of symptoms, categorizing symptoms as either major or minor. It also explores factors influencing healthcare-seeking propensity and examines whether healthcare-seeking propensity affects healthcare utilization and preventable hospitalizations.

Methods A cross-sectional study was conducted in 2023 in rural areas of Dangyang city, Hubei province, and Xishui county, Guizhou province. A sample of 780 patients, including those with hypertension, diabetes, or both, was obtained using multistage stratified cluster sampling. Healthcare-seeking propensity was evaluated using a questionnaire with two subscales for major and minor symptoms, which were analyzed separately. Multiple linear regression was used to identify factors influencing healthcare-seeking propensity, while logistic regression was employed to examine the relationship between healthcare-seeking propensity and healthcare utilization, as well as preventable hospitalizations.

Results Rural Chinese patients had an average healthcare-seeking propensity score of 5.43 ± 1.99 for major symptoms and 8.04 ± 1.66 for minor symptoms. Factors such as age, education level, income, accessibility to healthcare, and enrollment in family doctor services influenced healthcare-seeking propensity for both types of symptoms. Patients' self-rated health status and the type of disease affected their healthcare-seeking propensity for major symptoms, while primary healthcare service quality solely impacted the propensity for seeking care for minor symptoms. Logistic regression demonstrated that after accounting for confounding factors, a higher propensity for seeking care for minor symptoms was associated with increased healthcare service utilization (AOR = 1.402, 95% CI:

*Correspondence:
Shan Lu
shanlu@hust.edu.cn

Full list of author information is available at the end of the article

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1.259–1.562), whereas a higher propensity for major symptoms remarkably reduced preventable hospitalizations (AOR = 0.501, 95% CI: 0.428–0.587).

Conclusions Rural Chinese patients exhibited different healthcare-seeking propensity depending on the type of symptoms. The healthcare-seeking propensity in the western region was lower than that in the central region, highlighting regional disparities. Seeking care for minor symptoms results in more outpatient visits, whereas addressing major symptoms reduces preventable hospitalizations. Establishing a robust network of primary healthcare facilities and emphasizing the role of family doctors can boost patients' propensity to seek healthcare.

Clinical trial number Not applicable.

Keywords Healthcare-seeking propensity, Non-communicable diseases, Hypertension, Diabetes, Preventable hospitalizations, Healthcare utilization, Rural China

Introduction

In designing an efficient and cost-effective healthcare system, obtaining insights into the drivers of healthcare-seeking behavior and their subsequent impact on healthcare utilization is necessary. A critical facet of this endeavor is understanding the propensity to seek healthcare, which can vary among individuals, leading to variations in healthcare utilization, preventable hospitalizations and health outcomes [1–3].

Chronic diseases, also known as non-communicable diseases (NCDs), cause 41 million deaths annually, accounting for 74% of the total number of deaths worldwide [4]. In 2019, chronic diseases accounted for 1.62 billion disability adjusted life years (DALYs), accounting for 63.8% of the total global DALYs [5]. The situation in China is even more alarming, with chronic diseases contributing to 88.5% of total mortality in 2019 [6]. The burden of chronic diseases in China is particularly pronounced, with more than 500 million individuals affected by hypertension or diabetes, which together represent over 42% of the nation's overall chronic disease burden [5, 6]. Given the magnitude of this burden, addressing chronic diseases requires not only effective clinical interventions but also a focus on patient behaviors that influence health outcomes. Effective management of non-communicable diseases hinges on early detection, consistent screening, and prompt, suitable treatment. A crucial factor in achieving early detection and appropriate treatment is patients' propensity to promptly seek medical attention and actively collaborate with healthcare providers [5, 7–8]. Higher healthcare-seeking propensity not only improves health outcomes and quality of life but also enhances the efficiency and sustainability of healthcare systems [7, 8].

However, research on the healthcare-seeking propensity of patients with chronic conditions, particularly in rural China, remains limited [9–11]. Patients in China have the freedom to choose their initial healthcare provider [12], yet existing quantitative studies tend to focus on the factors that influence these choices, such as patient-related, provider-related, and contextual factors [9–11, 13]. On the contrary, studies conducted outside of China often focus on healthcare-seeking tendencies for specific illnesses, especially in the realm of mental health [14–16], which is a notable gap found in research addressing the healthcare-seeking propensity of individuals with chronic conditions such as hypertension or diabetes.

Anderson's behavioral model of healthcare utilization provides a comprehensive framework for understanding the factors that drive healthcare utilization [17]. The model identifies three key categories of factors: predisposing factors (such as age and gender), enabling factors (such as access to healthcare and socioeconomic status), and need factors (related to the perceived and evaluated health status) [18]. This study aims to apply Anderson's model to identify the underlying factors that influence the healthcare-seeking propensity of patients with hypertension or diabetes in rural China.

Therefore, this study primarily aims to evaluate the healthcare-seeking propensity among patients with hypertension or diabetes using a validated questionnaire. The secondary aim of this study is to investigate the factors influencing healthcare-seeking propensity in these patients by introducing the Anderson's behavioral model of healthcare utilization. Finally, the study seeks to examine the relationship between patients' healthcare-seeking propensity and healthcare utilization and the association between healthcare-seeking propensity and preventable hospitalizations. Our hypothesis is that patients with high healthcare-seeking propensity tend to exhibit increased healthcare utilization and low rates of preventable hospitalizations.

Method

Study design and participants

This cross-sectional study was performed in 2023 in Dangyang City (which was a county-level city), Hubei province, from central China and Xishui County, Guizhou Province, from western China. By the end of 2022, Xishui county had a resident population of 584,100 people. In 2022, the gross regional product of the county reached

US$3.90 billion (exchange rate in 2022: RMB¥6.73 to US$1.00), and the per capital gross regional product reached US$6,683.80. At the end of the year, the county had 485 healthcare facilities, 4,498 beds in healthcare facilities, and 4,639 healthcare providers. On the contrary, Dangyang City, also a county-level administrative unit (despite its name suggesting a higher level of administration) had a resident population of 418,500 people by the end of 2022. In 2022, the city achieved a gross domestic product of 9.34 billion dollars, with per capital gross regional product of US$22,317.80. The city had 265 healthcare facilities, with 2,810 beds. In Xishui and Dangyang, healthcare facilities primarily consisted of county hospitals, township health centers, and village clinics. Township health centers and village clinics provided primary care, and county hospitals and township health centers provided inpatient and outpatient services.

A multistage stratified cluster sampling was used to determine the participants of patients with hypertension in each county (Fig. 1). In the first stage, two townships that were relatively far away from the county center and two townships that were relatively close to the county center were randomly selected in each county. In the second stage, in each township, two villages that were relatively far away from the township center and two villages that were relatively close to the township center were randomly selected. In the third stage, 20 patients with hypertension were randomly selected in each village. All available patients with diabetes were exhaustively involved because of their relatively low number. Patients with hypertension or diabetes were identified using residents' health records maintained by local village doctors. This sampling method aimed to ensure geographic representativeness by including participants from diverse residential areas, avoiding over-concentration in any specific location.

The inclusion criteria were as follows: (1) study participants should be local residents who were over 18; (2) study participants must be able to understand the content of the questionnaire; (3) study participants were diagnosed with hypertension or diabetes.

In summary, 32 villages were identified in Dangyang and Xishui, and 800 patients (459 patients with hypertension, 155 patients with diabetes, 186 patients with both conditions) were invited to participate in the survey. 20 cases with missing data in the surveys or with wrong information were excluded. The final sample size was 780 (448 patients with hypertension, 153 patients with diabetes, 179 patients with both conditions) from 32 villages, with a response rate of 97.5% (780 out of 800).

Measurements of variables

Healthcare-seeking propensity measurement

In this study, healthcare-seeking propensity is the dependent variable, and we investigate several factors that may influence this propensity as independent variables. To evaluate healthcare-seeking propensity, we employed an established and validated questionnaire adapted from a previous study [8]. The questionnaire measures the participants' propensity to seek care based on the severity of symptoms, categorizing symptoms as either major or minor (Table 1). The questionnaire consists of 7 items related to major symptoms and 9 items related to minor symptoms. The items are rated on a 4-point Likert scale for major symptoms (ranging from 1 = not important to 4 = extremely important) while the items for minor symptoms are rated with a binary yes/no response format (1 = yes, 0 = no). A higher score indicates a higher propensity to seek care. The overall score is calculated as the sum of the scores from all items [8, 19]. To account for the differences in the measurement scales between the major and minor symptoms subscales, we normalized both subscale scores to a range of 0 to 10 using the formula:

$$\text{Normalized Score} = \frac{\text{Original Score} - \text{Minimum Score}}{\text{Maximum Score} - \text{Minimum Score}} \times 10$$

However, the minor symptoms subscale is based on binary yes/no responses, where a higher frequency of 'yes' answers automatically increases the score [20]. Since the scores for minor symptoms may be disproportionately influenced by the number of positive responses, direct comparisons between the normalized scores of the two subscales should be approached with caution. Such comparisons may not accurately reflect equivalent levels of propensity.

The Cronbach's alpha for the major symptoms subscale was 0.838, and for the minor symptoms subscale, it was 0.831, indicating good reliability for both subscales. The questionnaire, originally developed in English, was evaluated for use in Chinese populations through a pilot study conducted by our research team with a small sample from the target population. This pilot study ensured the clarity, understandability, and relevance of the questions, confirming the appropriateness of the scale for this research.

Factors influencing Healthcare-seeking propensity

The factors that influence healthcare-seeking propensity in this study are categorized into three groups: predisposing factors, enabling factors, and need factors based on Andersen's behavioral model [17]. These factors are described in Table 2. Each category includes variables that are hypothesized to influence the likelihood of an individual seeking healthcare services. The following

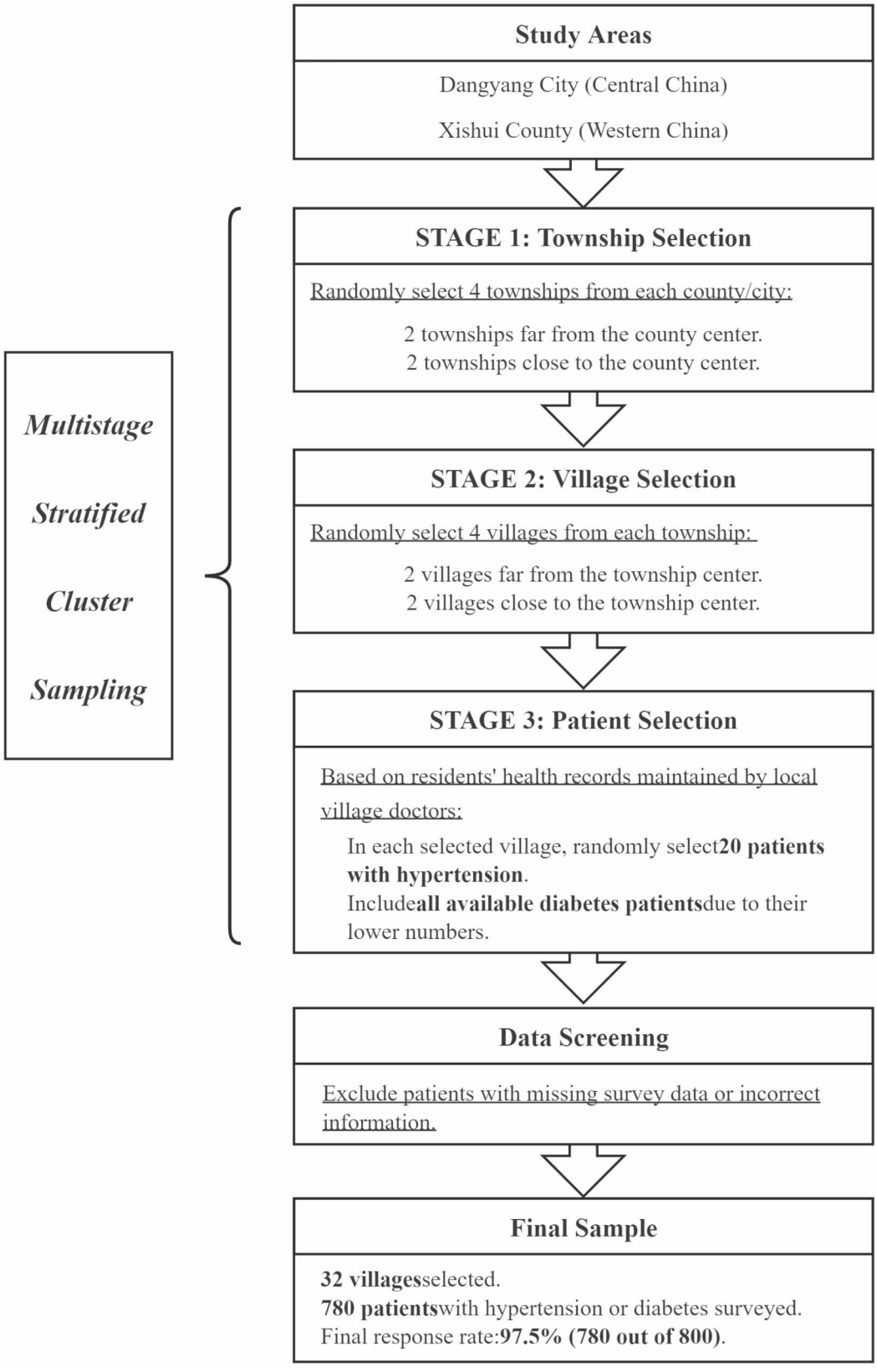

Fig. 1 Flow diagram of the sampling procedure for hypertension or diabetes patients

Table 1 Health-seeking propensity questionnaires based on two questions

Propensity to seek care (major symptoms)
How important would it be for you to see a doctor if you had…?
1) Weight loss of more than 2 kg in a month when not dieting
2) Shortness of breath with light exercise or light work
3) Chest pain when exercising
4) Loss of consciousness, fainting or passing out
5) Headache for more than one day
6) Abdominal pain for more than one day
7) Severe worries for more than a month
Answering categories: "*extremely important, rather important, somewhat important, and not important*".
Propensity to seek care (minor symptoms)
Do you expect to benefit from a medical visit for…?
1) Stomach problems
2) Shoulder and neck pain
3) Feeling nervous
4) Diarrhea
5) Sore throat
6) Headache
7) Feeling tired
8) Flu
9) Feeling nauseous
Answering categories were: *yes, no*

outlines how these factors were measured in the context of this study.

Predisposing factors refer to demographic and personal characteristics that may influence healthcare-seeking behavior. These include age, gender, education level, body mass index (BMI), and marital status. These variables were selected based on previous research that suggests they are important determinants of healthcare utilization.

Enabling factors include resources that facilitate or hinder an individual's access to healthcare. This category encompasses accessibility to healthcare facilities, family income, and the quality of primary healthcare services.

Need factors reflect individuals' health conditions and their perceived need for healthcare. These include the type of disease (hypertension, diabetes, or both) and self-rated health status, which are expected to influence an individual's decision to seek medical care.

To better illustrate the factors influencing healthcare-seeking propensity, Table 2 presents the definitions and categories of these factors.

Healthcare utilization and preventable hospitalizations

To further examine whether healthcare-seeking propensity affects healthcare utilization and preventable hospitalizations, we measured healthcare utilization and preventable hospitalizations as key indicators. Healthcare utilization was quantified by the number of outpatient visits made by participants within the span of last year, categorized into two groups: fewer than four visits and four or more visits [28]. The Agency for Healthcare Research and Quality (AHRQ) defines potentially preventable hospitalizations as hospital admissions for conditions such as heart failure, diabetes, hypertension, and asthma, where effective primary care could have prevented the need for hospitalization [29, 30]. In this study, preventable hospitalizations were identified through participant surveys and consultations with village doctors, who provided information on hospitalizations related to hypertension or diabetes in the past year. The criteria for defining preventable hospitalizations were based on the OECD's inclusion and exclusion guidelines. The inclusion criteria were as follows: age ≥ 18 years; main diagnosis, hypertension or diabetes. The exclusion criteria were as follows: maternal cases, cases of death in hospital during admission, cases referred from other institutions, and patients admitted and discharged on the same day.

Statistical analysis

Multiple linear regression analysis was conducted to identify the factors associated with healthcare-seeking propensity for major and minor symptoms among patients with hypertension or diabetes, as it allows for the analysis of relationships between continuous dependent variables and multiple independent predictors. Separate analyses were conducted for major and minor symptoms. The assumptions for multiple linear regression were tested separately for the major symptoms and minor symptoms models prior to model fitting. Linearity was supported by scatterplots. Normality of residuals was evaluated with Kolmogorov-Smirnov (K-S) tests. For the major symptoms model, the K-S test resulted in $D = 0.038$, $p = 0.11$, indicating normality. Similarly, the minor symptoms model showed $D = 0.042$, $p = 0.09$, also suggesting normality. Homoscedasticity was examined via residual plots, revealing evenly distributed residuals around zero in both models, confirming constant variance. Independence of residuals was assessed using Durbin-Watson (DW) statistics. The major symptoms model had a DW value of 1.679, while the minor symptoms model had a DW value of 1.523, indicating no significant autocorrelation. Variance inflation factors (VIFs) for all predictors were below 10, confirming the absence of multicollinearity. These diagnostic tests support the validity of both multiple linear regression models.

should I do this??

the closer to 0, the more normal?

In addition, logistic regression models were used to investigate the impact of healthcare-seeking propensity on outpatient visits and preventable hospitalization for major and minor symptoms. Logistic regression models were employed because the dependent variables, outpatient visits and preventable hospitalizations, are binary outcomes. This method allows us to assess the impact of healthcare-seeking propensity while controlling for confounding variables such as sociodemographic characteristics, disease characteristics, and environmental factors [18.19]. The assumptions underlying logistic regression

Table 2 Definitions and categories of the associated factors

Factors	Definitions and categories
Predisposing Factors	
Age	Age of the participants 0 = middle-aged individuals (< 60 years), 1 = older adults (≥ 60 years)
Gender	Gender of the participants 0 = male,1 = female
Educational attainment	Education level of the participants 0 = primary education and below, 1 = secondary education, 2 = higher education
Body mass index	BMI of the participants is categorized into three groups based on World Health Organization (WHO) BMI classifications 0 = BMI ≤ 18.5(underweight), 1 = BMI 18.5–23.9 (normal weight), 2 = BMI>23.9 (overweight)
Marital status	Marital status of the participants 0 = married, 1 = others
Enabling factors	
Accessibility to healthcare facilities	The amount of time required to reach the nearest healthcare facility from the participant's residence using the most commonly used or easily accessible mode of transportation, including walking, cycling, riding motorcycles (if available), or using public transportation or private vehicles, depending on what was most accessible to the individual participants 0 = near (≤ 15 min), 1 = moderate (15–30 min), 2 = far (>30 min)
Family income	Annual family income of the participants, using the average value of the household income of all study subjects 4795$ as the threshold, incomes above this value are considered "above average." 0 = below average, 1 = above average
Primary healthcare service quality	The quality of healthcare service accessed through PCAT scales [21–24], a score exceeding 60% on the scale was defined as "high," whereas a score falling below 60% was categorized as "low." "0 = High (≥ 60%), 1 = low (< 60%)
Family doctor contracted service[a]	The contracted status of the participants 0 = Contracted, 1 = Uncontracted
Need factors	
Disease type	Disease type of the participants 0 = Hypertension, 1 = Diabetes, 2 = Both
Self-rated health	Self-rated health of the participants 0 = rather bad, 1 = bad, 2 = normal,3 = good,4 = rather good

[a]"Family Doctor Contracted service" refers to China's Family Doctor Contracted Service Policy enacted in 2016, this initiative fosters continuity of care by urging individuals to forge a steadfast relationship with a designated primary care provider [25–27]

were found to be adequately met, and the goodness of fit for the models were calculated and presented in the **Results**.

Results

Study participants

Table 3 provides a detailed overview of the characteristics of the study participants. In total, 780 participants were included in the study. Of the participants, 564 (72.4%) were 60 years old or older. The gender distribution was 333 men (42.7%) and 447 women (57.3%). Only 30.8% of participants had an experience of secondary (29.5%) or higher education. With regard to income, 462 participants (59.2%) reported an income less than 4795$. Regarding accessibility, 64.7% and 8.3% of participants took 15–30 min and more than 30 min to reach the nearest healthcare facility, respectively. These time estimates refer to the time spent using various modes of transportation, including walking, cycling, riding motorcycles (if available), or using other means such as public transportation or private vehicles, depending on what was accessible to the individual participants. The evaluation of primary healthcare service quality revealed that 452 participants (57.9%) rated it as high (score ≥ 3), whereas 328 participants (42.1%) considered it low (score < 3). Less than half of the participants (42.9%) had contracted family doctor services. 448 participants (57.4%) had hypertension, 153 participants (19.6%) had diabetes, and 179 participants (22.9%) had both conditions. Self-rated health indicated that 35.5% of participants perceived their health as bad or rather bad, whereas 31.8% of participants perceived their health as good or rather good.

Healthcare-seeking propensity scores

Among the participants, the average score of healthcare-seeking propensity for major symptoms was 5.43 ± 1.99. A loss of consciousness score of 7.38 ± 3.02, with a chest pain when exercising score of 7.22 ± 3.11, indicated that patients frequently agreed on the necessity of a doctor's visit. Weight loss of more than 2 kg in a month when not dieting with a score of 3.56 ± 2.02 has the least scores. The propensity to seek healthcare for severe symptoms in the

Table 3 Characteristics of participants

Variable	Category	*n*	Percentage
Age	< 60	216	27.6
	≥ 60	564	72.4
Gender	Men	333	42.7
	Women	447	57.3
BMI	< 18.5	59	7.6
	18.5 ~ 23.9	374	47.9
	≥ 23.9	339	43.7
Marital status	Married	677	86.8
	Others	103	13.2
Education Attainment	Primary education and below	540	69.1
	Secondary education	230	29.5
	Higher education	10	1.3
Family Income($)	< 4795	462	59.2
	≥ 4795	251	32.1
Accessibility	near	208	26.6
	moderate	505	64.7
	far	65	8.3
Primary Healthcare Service Quality	High	452	57.9
	Low	328	42.1
Family Doctor Contracted service	Contracted	445	42.9
	Uncontracted	335	57.1
Type of Disease	Hypertension	448	57.4
	Diabetes	153	19.6
	Both	179	22.9
Self-rated Health	Rather good	82	10.5
	Good	166	21.3
	normal	255	32.7
	bad	199	25.5
	Rather bad	78	10.0

central region (5.47 ± 2.03) is higher than that in the western region (5.39 ± 1.96).

The average score of healthcare-seeking propensity for minor symptoms was 8.04 ± 1.66. Patients expected to benefit most from seeking healthcare for stomach problems, which scores 9.37 ± 1.66. Feeling nervous with a score of 7.35 ± 2.50 has the least scores. The propensity to seek healthcare for minor symptoms in the central region with a score of 8.17 ± 1.72 is higher than that in the western region (7.91 ± 1.59). Detailed scores for each symptoms are presented in Table 4.

Table 4 Healthcare-seeking propensity scores

Symptom	Overall patients (*n* = 780)	Western Region (*n* = 402)	Central Region (*n* = 378)
Major symptoms			
Chest Pain	7.22 ± 3.11	7.05 ± 3.13	7.41 ± 3.07
Short of Breath	5.59 ± 3.14	5.58 ± 3.18	5.60 ± 3.09
Severe worries (> 1 month)	4.65 ± 2.66	4.72 ± 2.74	4.57 ± 2.58
Weight Loss(> 2Kg)	3.56 ± 2.02	3.46 ± 1.84	3.66 ± 2.00
Headache(> 1d)	4.62 ± 2.65	4.67 ± 2.70	4.56 ± 2.60
Abdominal Pain(> 1d)	4.98 ± 2.84	5.13 ± 2.98	4.83 ± 2.67
Loss of Conscious	7.38 ± 3.02	7.13 ± 3.15	7.65 ± 2.85
Total Scores	**5.43 ± 1.99**	**5.39 ± 1.96**	**5.47 ± 2.03**
Minor symptoms			
Stomach problems	9.37 ± 1.66	9.32 ± 1.72	9.43 ± 1.59
Shoulder and neck pain	8.65 ± 2.22	8.55 ± 2.27	8.77 ± 2.16
Feeling nervous	7.35 ± 2.50	7.39 ± 2.50	7.30 ± 2.50
Diarrhea	9.22 ± 1.81	9.23 ± 1.81	9.22 ± 1.82
Sore throat	8.99 ± 2.01	9.03 ± 1.98	8.96 ± 2.04
Headache	9.32 ± 1.71	9.40 ± 1.62	9.23 ± 1.80
Feeling tired	7.78 ± 2.49	7.62 ± 2.50	7.95 ± 2.46
Flu	9.32 ± 1.71	9.30 ± 1.73	9.34 ± 1.70
Feeling nauseous	8.81 ± 2.13	8.72 ± 2.19	8.92 ± 2.06
Total Scores	**8.04 ± 1.66**	**7.91 ± 1.59**	**8.17 ± 1.72**

Factors associated with healthcare-seeking propensity

Tables 5 and 6 present the factors associated with healthcare-seeking propensity for major and minor symptoms, respectively. For major symptoms, the overall fit of the model was statistically significant (F = 17.169, $p < 0.001$), indicating that the model explained a substantial portion of the variance in healthcare-seeking propensity. The adjusted R^2 value was 0.462, suggesting that our model accounted for approximately 46.2% of the variance in healthcare-seeking propensity, highlighting the significant influence of the included predictors. Age significantly influenced healthcare-seeking behavior, with participants aged 60 years or above demonstrating a higher propensity compared to participants below 60 years of age. (β = 0.674, 95% CI: 0.339–1.008, $p < 0.001$). Education attainment was also significant; individuals with secondary education showed a greater tendency to seek healthcare than those with primary education or below (β = 0.585, 95% CI: 0.265–0.904, $p < 0.001$).Income played a crucial role, with above-average earners exhibiting a greater propensity to seek healthcare (β = 0.896, 95% CI: 0.588–1.205, $p < 0.001$). Regarding self-rated health, individuals rating their health as "bad" were significantly more likely to seek care than those rating it as "rather bad" (β = 1.292, 95% CI: 0.675–1.970, $p < 0.001$).Type of disease was a significant predictor, with individuals with diabetes having a higher propensity to seek healthcare than those with hypertension (β = 0.419, 95% CI: 0.015–0.822, $p = 0.042$). However, individuals with both diabetes and hypertension did not show a significant difference in healthcare-seeking propensity (β = 0.131, 95% CI: -0.243–0.505, $p = 0.491$). Accessibility to healthcare services was also critical; individuals with moderate accessibility (15–30 min) had a significantly lower propensity to seek healthcare compared to those with short accessibility (< 15 min; β=–0.660, 95% CI: -0.993–-0.326, $p < 0.001$). Those with far accessibility (> 30 min) exhibited an

Table 5 Factors associated with healthcare-seeking propensity for major symptoms

Variables	β(95%CI)	S.E	t	*P*
(Constant)	4.346 (3.561 ~ 5.131)	0.400	10.868	< 0.001
AGE (with age<60 as the reference)				
Older adults (≥ 60 years)	0.674 (0.339 ~ 1.008)	0.170	3.951	< 0.001
Gender (with male as the reference)				
Female	-0.026 (-0.325 ~ 0.274)	0.152	-0.168	0.867
BMI (with BMI ≤ 18.5 as the reference)				
BMI 18.5–23.9	0.149 (-0.409 ~ 0.707)	0.284	0.524	0.600
BMI>23.9	0.271 (-0.295 ~ 0.838)	0.288	0.941	0.347
Marital status (with married as the reference)				
Others	-0.402 (-0.821 ~ 0.017)	0.213	-1.886	0.060
Education Attainment (with primary education and below as the reference)				
Secondary education	0.585 (0.265 ~ 0.904)	0.163	3.592	< 0.001
Higher education	0.197 (0.976 ~ 1.369)	0.597	0.329	0.742
Income(with below average as the reference)				
Above average	0.896 (0.588 ~ 1.205)	0.157	5.703	< 0.001
Self-rated Health(with rather bad as the reference)				
Bad	1.292 (0.675 ~ 1.970)	0.315	4.108	< 0.001
Normal	1.040 (0.527 ~ 1.554)	0.261	3.982	< 0.001
Good	0.591 (0.095 ~ 1.087)	0.253	2.338	0.020
Rather good	1.016 (0.495 ~ 1.537)	0.265	3.827	< 0.001
Type of Disease(with hypertension as the reference)				
Diabetes	0.419 (0.015 ~ 0.822)	0.206	2.036	0.042
Both	0.131 (-0.243 ~ 0.505)	0.190	0.690	0.491
Accessibility(with near ≤ 15 min as the reference)				
Moderate (15–30 min)	-0.660 (-0.993~-0.326)	0.170	-3.885	< 0.001
Far (>30 min)	-0.921 (-1.442~-0.400)	0.265	-3.471	0.001
Primary Healthcare Service Quality(with High quality as the reference)				
low (< 60%)	-0.065 (-0.356 ~ 0.225)	0.148	-0.442	0.659
Family Doctor Contracted service(with Contracted as the reference)				
Uncontracted	-0.620 (-0.909~-0.331)	0.147	-4.217	< 0.001

even greater decrease in propensity (β=−0.921, 95% CI: -1.442–-0.400, $p = 0.001$). Individuals receiving uncontracted family doctor services had a lower propensity to seek healthcare compared to those with contracted services (β=−0.620, 95% CI: -0.909–-0.331, $p < 0.001$).

For minor symptoms, the overall fit of the model was statistically significant (F = 13.000, $p < 0.001$), indicating that the model explained a substantial portion of the variance in healthcare-seeking propensity. The adjusted R^2 value was 0.413, suggesting that our model accounted for approximately 41.3% of the variance in healthcare-seeking propensity, highlighting the significant influence of the included predictors. Age significantly influenced healthcare-seeking behavior, with older adults (≥ 60 years) demonstrating a higher propensity compared to younger individuals (β = 0.193, 95% CI: 0.061–0.447, $p < 0.001$). education attainment significantly impacted healthcare-seeking, with secondary (β = 0.485, 95% CI: 0.165–0.804, $p = 0.001$) and higher education (β = 1.121, 95% CI: 0.235–2.007, $p = 0.013$) showing greater propensity. Income also played a role, with above-average earners more likely to seek care (β = 0.364, 95% CI: 0.130–0.597, $p = 0.002$).Accessibility was crucial, with moderate (β=−0.400, 95% CI: -0.654–-0.146, $p = 0.002$) and far distances (β=−0.512, 95% CI: -0.817–-0.207, $p < 0.001$) reducing the likelihood of seeking care. Lower primary healthcare quality (β=−0.458, 95% CI: -0.680–-0.237, $p < 0.001$) and lack of contracted family doctor services (β=−0.365, 95% CI: -0.584–-0.146, $p = 0.008$) also decreased healthcare-seeking propensity.

Relationship between healthcare-seeking propensity and healthcare utilization

The empirical findings presented in Table 7 demonstrate divergent effects of healthcare-seeking propensity on healthcare utilization concerning major and minor symptoms. Upon controlling for confounding variables, our analysis revealed that the propensity for major symptoms did not exert a statistically significant influence on healthcare utilization. Conversely, the propensity for minor symptoms exhibited a notable impact on

Table 6 Factors associated with healthcare-seeking propensity for minor symptoms

Variables	β(95%CI)	S.E	t	*P*
(Constant)	9.420 (8.826 ~ 10.014)	0.303	31.128	< 0.001
AGE (with age<60 as the reference)				
Older adults (≥ 60 years)	0.193 (0.061 ~ 0.447)	0.129	3.094	< 0.001
Gender (with male as the reference)				
Female	-0.122 (-0.348 ~ 0.105)	0.115	-1.054	0.292
BMI (with BMI ≤ 18.5 as the reference)				
BMI 18.5–23.9	0.010 (-0.412 ~ 0.432)	0.215	0.046	0.964
BMI>23.9	0.054 (-0.374 ~ 0.483)	0.218	0.249	0.803
Marital status (with married as the reference)				
Others	0.137 (-0.180 ~ 0.454)	0.161	0.850	0.395
Education Attainment (with primary education and below as the reference)				
Secondary education	0.485 (0.165 ~ 0.804)	0.163	3.091	< 0.001
Higher education	1.121 (0.235 ~ 2.007)	0.451	2.485	0.013
Income(with below average as the reference)				
Above average	0.364 (0.130 ~ 0.597)	0.119	3.057	0.002
Self-rated Health(with rather bad as the reference)				
Bad	-0.222 (-0.689 ~ 0.244)	0.238	-0.935	0.350
Normal	-0.047 (-0.435 ~ 0.340)	0.197	-0.241	0.810
Good	-0.026 (-0.402 ~ 0.349)	0.191	-0.139	0.890
Rather good	-0.111 (-0.506 ~ 0.284)	0.201	-0.551	0.582
Type of Disease(with hypertension as the reference)				
Diabetes	-0.256 (-0.561 ~ 0.050)	0.156	-1.643	0.101
Both	-0.031 (-0.314 ~ 0.252)	0.144	-0.212	0.832
Accessibility(with near ≤ 15 min as the reference)				
Moderate (15–30 min)	-0.400 (-0.654~-0.146)	0.129	-3.091	0.002
Far (>30 min)	-0.512 (-0.817~-0.207)	0.155	-3.295	< 0.001
Primary Healthcare Service Quality(with High quality as the reference)				
low (< 60%)	-0.458 (-0.680~-0.237)	0.113	-4.060	< 0.001
Family Doctor Contracted service(with Contracted as the reference)				
Uncontracted	-0.365 (-0.584~-0.146)	0.111	-3.380	0.008

Table 7 Impact of Healthcare seeking propensity for major symptoms and minor symptoms on Healthcare utilization

Factors	Healthcare Utilization	
	Odds ratio adjusted(95%CI)	*P*
Propensity for major symptoms	1.000—1.185	0.051
Propensity for minor symptoms	1.259—1.562	< 0.001

Table 8 Impact of Healthcare seeking propensity for major symptoms and minor symptoms on preventable hospitalization

Factors	Preventable Hospitalization	
	Odds ratio adjusted(95%CI)	*P*
Propensity for major symptoms	0.428—0.587	< 0.001
Propensity for minor symptoms	0.77—2.077	0.061

healthcare utilization ($P < 0.001$, AOR = 1.402, 95% CI: 1.259–1.562) after adjusting for confounding factors.

Relationship between healthcare-seeking propensity scores and preventable hospitalizations

The empirical examination presented in Table 8 illuminates the divergent effects of healthcare-seeking propensity on preventable hospitalizations concerning minor and major symptoms. Following adjustments for confounding variables, our analysis revealed no statistically significant association between healthcare-seeking propensity for minor symptoms and preventable hospitalization ($P = 0.061$, AOR = 0.1.512, 95% CI: 0.77–2.077). By contrast, the propensity for major symptoms exhibited a notable impact on preventable hospitalization ($P < 0.001$, AOR = 0.501, 95% CI: 0.428–0.587).

Discussion

Our findings showed that the propensity to seek care for both major and minor symptoms was significantly lower in the western region compared to the central region,

highlighting notable regional disparities in healthcare-seeking propensity among rural patients. Age, education level, income, accessibility to healthcare facilities, and participation in family doctor contract services were found to influence healthcare-seeking propensity for major and minor symptoms. On the contrary, certain factors, such as self-rated health status and type of disease, only affected healthcare-seeking propensity for major symptoms, and primary healthcare service quality solely impacted healthcare-seeking propensity for minor symptoms. A higher propensity to seek care was associated with more outpatient visits. Conversely, propensity to seek care for major symptoms was linked to preventable hospitalizations.

The central area demonstrates slightly higher healthcare-seeking scores for major and minor symptoms compared with the western region because of demographic and economic disparities. Central rural areas, with an older population, higher income, and better education [31], tend to exhibit a higher propensity to seek medical services [32]. In addition, healthcare accessibility in the central region is better, with shorter travel times to medical facilities (15.07 min vs. 21.03 min), encouraging proactive approach to healthcare [33].

To identify the factors influencing patients' propensity to seek care, the Andersen's behavioral model of healthcare utilization was applied. The factors influencing patients' propensity to seek medical care for major symptoms and minor symptoms largely overlap, with age, education level, income, accessibility to healthcare, and enrollment in family doctor contract services playing important roles. These findings are consistent with prior research, indicating that patients with greater healthcare needs and higher income, educational attainment exhibit an increased propensity to seek medical care [8, 19, 34–37]. Patients with improved access, continuity in primary healthcare facilities, and effective communication with family doctors exhibited a higher propensity to seek care [38–40]. Based on the findings of our study and prior research, family doctor services play a pivotal role in the management of chronic diseases in rural settings [41]. The management of conditions such as hypertension and diabetes by family physicians has demonstrated efficacy in regulating blood pressure and glucose levels [42], augmenting patients' healthcare-seeking inclinations and health literacy, and mitigating preventable hospitalization, thereby advancing the overall quality of life [24, 40, 43]. The results also indicated that type of disease and self-rated health status only affect seeking care for major symptoms, whereas primary healthcare service quality only impacts seeking care for minor symptoms. The type of disease significantly influences the propensity to seek care for major symptoms, with diabetic patients demonstrating a greater inclination to do so compared to hypertensive patients. This may be attributed to higher prevalence of hypertension, leading many patients to adapt and regard it as a normal aspect of life, thereby reducing their urgency to seek medical attention for significant symptoms. In addition, diabetic patients typically require more strict self-management and regular monitoring, whereas hypertensive patients often adopt a more relaxed approach to their health management. Regarding self-rated health status, individuals who perceive their health as good may be more attentive and sensitive to the occurrence of major symptoms, leading them to seek medical care more promptly [44]. In contrast, for minor symptoms, they may choose to wait and observe or self-manage. The fact that primary healthcare service quality solely impacts healthcare-seeking propensity for minor symptoms could be attributed to the severity of symptoms influencing individuals' preferences for different levels of medical institutions. When symptoms are severe, most patients tend to avoid seeking care at primary healthcare facilities and instead opt for higher tier comprehensive hospitals [11, 33, 35, 45]. Therefore, primary healthcare service quality does not exert an impact on the propensity to seek medical care for major symptoms.

Furthermore, whether patients' healthcare-seeking propensity impacts healthcare utilization and preventable hospitalization was explored. Our hypothesis that higher scores for the propensity to seek care would lead to more outpatient visits but lower rates of preventable hospitalization was partially supported by the results. A higher propensity to seek care was correlated with more outpatient visits, but this effect was not discernible for severe symptoms, which is consistent with the findings of other studies [8, 46].

Regarding preventable hospitalization, our study found an association with the propensity to seek care for major symptoms, which contradicts some previous research [8, 46]. These earlier studies reported that the majority of avoidable hospitalizations were not significantly linked to healthcare-seeking propensity. However, another study [47] observed that individuals in areas with high rates of preventable hospitalizations tended to delay outpatient care longer than those in areas with low rates, which aligns with our findings. A literature review also highlighted that strong primary care, characterized by adequate physician supply and long-term patient-provider relationships, reduces the risk of avoidable hospitalizations [39]. The relationship between healthcare-seeking propensity for major symptoms and preventable hospitalization likely stems from the timely interventions that early medical attention provides, potentially preventing the worsening of conditions and reducing the need for hospitalization [48, 49].

The differential impact of healthcare-seeking propensity for major and minor symptoms on healthcare utilization and preventable hospitalization provides crucial insights at theoretical and practical levels. On the theoretical front, this study elucidates the multifaceted nature of healthcare-seeking propensity, particularly on discerning between major and minor symptoms. In the past, we typically assumed that a higher healthcare-seeking propensity always leads to higher service utilization, lower preventable hospitalizations. However, our findings show a different pattern. The relationship between healthcare-seeking propensity and service utilization varies depending on the type of symptom, with the impact differing between major and minor symptoms. In addition, the severity of symptoms exerts varying impacts on healthcare utilization. On the practical front, differentiating between minor and major symptoms leads to a more accurate measurement of healthcare-seeking propensity.

Limitation.

One limitation of our study is the lack of detailed analysis regarding the classification of participants based on their disease types, particularly the interplay between diabetes and hypertension. While hypertension and diabetes are known to be closely linked, with diabetes increasing the risk of hypertension [50], our dataset did not provide enough granularity to differentiate between individuals whose hypertension is secondary to diabetes and those with primary hypertension. As a result, we were unable to categorize participants based on the specific nature of their conditions, which may have impacted our understanding of how these diseases influence healthcare-seeking behavior. Future studies with more detailed clinical data, which include information on whether hypertension is primarily or secondarily related to diabetes, would be beneficial for a more nuanced analysis. Additionally, the categorization of participants based on their specific disease combinations could provide deeper insights into how comorbidities influence healthcare-seeking propensity.

Conclusion

This study sheds light on the healthcare-seeking propensity among rural patients with hypertension or diabetes in China, emphasizing the distinctions between major and minor symptoms. Factors influencing healthcare-seeking propensity include demographic, economic, and healthcare system characteristics. Patients with greater healthcare needs, higher income levels, and advanced educational attainment, coupled with enhanced accessibility to healthcare services, favorable experiences with primary healthcare delivery, and engagement with a designated family doctor, exhibit an elevated propensity to seek healthcare. A higher propensity to seek care was associated with increased outpatient visits, particularly for minor symptoms. Preventable hospitalizations were linked to healthcare-seeking propensity for major symptoms, underscoring the importance of timely medical intervention.

Given these findings, policy interventions should prioritize enhancing healthcare accessibility and strengthening primary healthcare services, particularly in rural areas. Establishing a robust network of township health centers and emphasizing the role of family doctors in delivering continuous care can foster patient trust and satisfaction with primary care, thereby boosting patients' propensity to seek healthcare. In addition, implementing initiatives to promote health literacy and awareness about the importance of early intervention for major and minor symptoms can further encourage timely healthcare-seeking behavior among patients with hypertension or diabetes.

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Author contributions
SL, YH designed research. SL, YH completed Writing-review & editing. SL, YH, WL, LC and JA collected data. YH, SL conducted data analysis. YH, WL performed data visualization. SL, YH Write original draft. WL conducted data curation. YH and SL completed review & editing.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate
The study was approved by the Ethics Committee of Tongji Medical College, Huazhong University of Science and Technology, with the ethics approval number IORG0003571. All methods were performed in accordance with relevant guidelines and regulations. This study adhered to the principles of the Declaration of Helsinki. Informed consent was obtained from all participants prior to their inclusion in the study.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]School of Medicine and Health Management, Tongji Medical College, Huazhong University of Science and Technology, No.13, Hangkong Road, Qiaokou District, Wuhan City 430030, China
[2]School of Basic Medicine, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China
[3]Research Centre for Rural Health Service, Key Research Institute of Humanities and Social Sciences of Hubei Provincial Department of Education, Wuhan, China

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